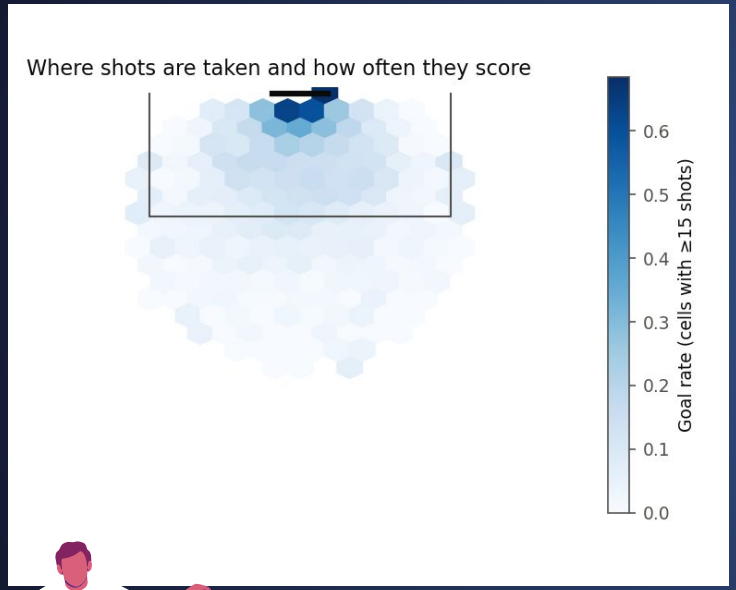
How much do body part & play pattern add to xG?

IS507 Midterm Project Presentation

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Why XG?



- Betting sites and media publish win probabilities built from statistical models
- We can't build a full match model, but we can study one key building block: expected goals (xG)
- xG = probability that a single shot becomes a goal
- Teams and players are judged by *summed* xG → if the probabilities are biased, every total is biased
- Goal: find which simple shot features add accuracy, and check if the probabilities can be trusted



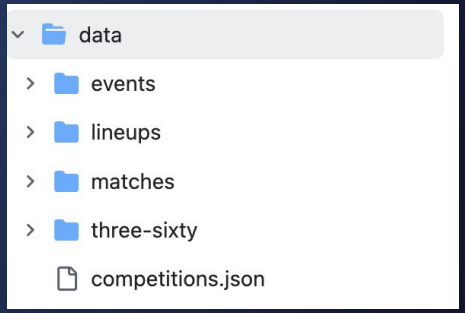
Research Question

How much do body part and play pattern improve a location-based xG model, and are its probabilities trustworthy?

	What	Detail
Y	is_goal	1 = goal, 0 = no goal · (only 9.5% are goals)
X	Shot features	distance & angle (from x, y) · body part · play pattern
Model	Logistic regression	$p = \frac{1}{1 + e^{-(b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k)}}$
Loss	Log loss	punishes confident wrong predictions; rewards honest probabilities

Data Sources

- **StatsBomb Open Data** (main)
  - Free, event-level data; 2015/16 = only season with complete top leagues
  - 4 leagues used: Premier League, La Liga, Serie A, Ligue 1
  - Folders: competitions.json · **matches**/ · **events**/ · lineups/ · three-sixty/ (we use the bold ones)
- **Football-Data.co.uk** (supporting)
  - One row per match · pre-match Bet365 odds → team strength
- Bundesliga excluded: source holds only 34 of 306 matches



What the Data Look Like

We built one shot table from 5.3 million raw events.

Dataset	One row =	Rows	Columns	Types
Raw events	one on-ball action	5,321,459	120	text, boolean, numeric, nested
Match list	one match	1,517	55	text, numeric
Shot table	one shot	37,888 (37,488 non-penalty)	20	text, numeric, boolean
Match odds	one match	1,520	65	numeric odds, text

```
Match files: 1,517
Total events (rows): 5,321,459
Events per match: mean 3,508, min 2,614, max 4,632
Distinct columns across all files: 120
Shots: 37,888 (0.7% of all events)

Column types in one example match:
text / category       35
boolean               34
numeric               16
nested (list/dict)     8

Most common event types (% of all events):
type
Pass             27.8
Ball Receipt*    26.0
Carry            21.0
Pressure          9.2
Ball Recovery     3.1
Duel              2.4
Clearance         1.4
Block             1.1
Dribble           1.0
Goal Keeper       0.9
```

One Row = One Shot

- **Features:** x, y → distance & angle · shot_body_part · play_pattern
- **Target:** shot_outcome → Goal = 1, everything else = 0
- **Processing:** shot_type (remove penalties) · match_id (split by match)
- **Reference only:** shot_statsbomb_xg (never a feature)

One Row = One Shot

1) RAW STATSBOMB EVENT (one shot, nested JSON)


```
{
  "minute": 4,
  "second": 18,
  "type": {
    "id": 16,
    "name": "Shot"
  },
  "play_pattern": {
    "id": 6,
    "name": "From Counter"
  },
  "team": {
    "id": 28,
    "name": "AFC Bournemouth"
  },
  "player": {
    "id": 3346,
    "name": "Joshua King"
  },
  "location": [
    100.0,
    27.9
  ],
  "shot": {
    "statsbomb_xg": 0.03997139,
    "end_location": [
      117.6,
      41.9,
      1.6
    ],
    "key_pass_id": "ec40b213-1cb6-4cf0-9921-8077d230ee70",
    "technique": {
      "id": 93,
      "name": "Normal"
    },
    "body_part": {
      "id": 40,
      "name": "Right Foot"
    },
    "type": {
      "id": 87,
      "name": "Open Play"
    },
    "outcome": {
      "id": 100,
      "name": "Saved"
    },
    "freeze_frame": [
      {
        "location": [
          99.0,
          34.7
        ],
        "player": {
          "id": 3713,
          "name": "Calum Chambers"
        },
        "position": {
          "id": 13,
          "name": "Right Center Midfield"
        },
        "teammate": false
      },
      {
        "location": [
          103.0,
          43.1
        ],
        "player": {
          "id": 3528,
          "name": "Kieran Gibbs"
        },
        "position": {
          "id": 6,
          "name": "Left Back"
        },
        "teammate": false
      },
      "... 7 more players ..."
    ]
  }
}
```


2) OUR SHOT TABLE (one row = one shot)

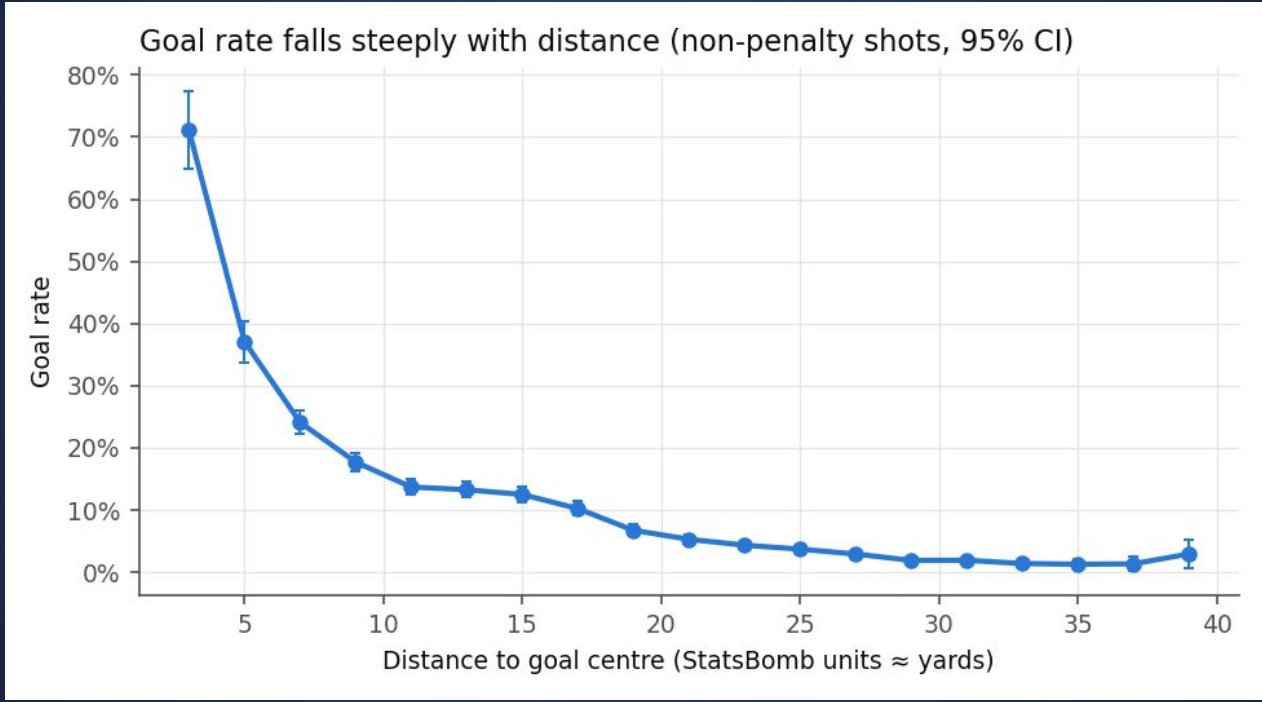
match_id	team	player	x	y
3754173	AFC Bournemouth	Joshua King	100.0	27.9
3754173	AFC Bournemouth	Harry Arter	103.7	42.7
3754173	Arsenal	Gabriel Armando de Abreu	112.4	38.0
3754173	Arsenal	Theo Walcott	118.5	50.8
3754173	Arsenal	Gabriel Armando de Abreu	113.6	48.5

shot_body_part	play_pattern	shot_type	shot_outcome	shot_statsbomb_xg
Right Foot	From Counter	Open Play	Saved	0.040
Right Foot	Regular Play	Open Play	Blocked	0.060
Head	From Corner	Open Play	Goal	0.144
Right Foot	From Throw In	Open Play	Blocked	0.014
Head	From Corner	Open Play	Post	0.058

What the Data Say

- Goal rate drops steeply with distance → location is the strongest signal → **Baseline 1 = distance + angle**
- Only **9.5%** of non-penalty shots are goals (3,569 / 37,488)
  - Predicting “no goal” every time = 90.5% accurate → accuracy is useless
  - We use log loss and calibration instead
- Headers come mostly from close range → raw rates mislead; need a model that holds location fixed

What the Data Say



Challenges We Found

Cleaning the data was a big part of the work

Challenge	Evidence	What we did
Nested JSON	Locations, freeze frames are nested lists	Flattened shots into our own table
Bundesliga incomplete	34 / 306 matches, all Leverkusen	Excluded
Penalties differ	75.0% scored vs 9.5%	Removed
Blanks that mean "False"	under_pressure 74.1%, shot_first_time 68.7% blank	Rule: blank = False (not used in current models)
Team names differ	45 / 98 names (e.g. "Man United")	Manual name map

Challenges We Found

team_name_map

fd_name	sb_name	league	note
Man United	Manchester United	Premier League	abbreviation
Man City	Manchester City	Premier League	abbreviation
Tottenham	Tottenham Hotspur	Premier League	short name
Newcastle	Newcastle United	Premier League	short name
West Ham	West Ham United	Premier League	short name
West Brom	West Bromwich Albion	Premier League	abbreviation
Leicester	Leicester City	Premier League	short name
Norwich	Norwich City	Premier League	short name
Swansea	Swansea City	Premier League	short name
Stoke	Stoke City	Premier League	short name
Bournemouth	AFC Bournemouth	Premier League	short name

```
=== Downloaded match list ===
Matches: 34
Date range: 2015-08-15 to 2016-05-14
Distinct teams: 18
Teams appearing most often:
Bayer Leverkusen                34
Borussia Mönchengladbach         2
Bayern Munich                    2
Borussia Dortmund                2
Werder Bremen                    2
Raw event files on disk: 34
```

Plan: Models

- Logistic regression, one feature group at a time:
  - **B0:** predict 9.3% for every shot (the floor)
  - **B1:** distance + angle (location only)
  - **M2:** + body part (foot = reference)
  - **M3:** + play pattern (regular play = reference)

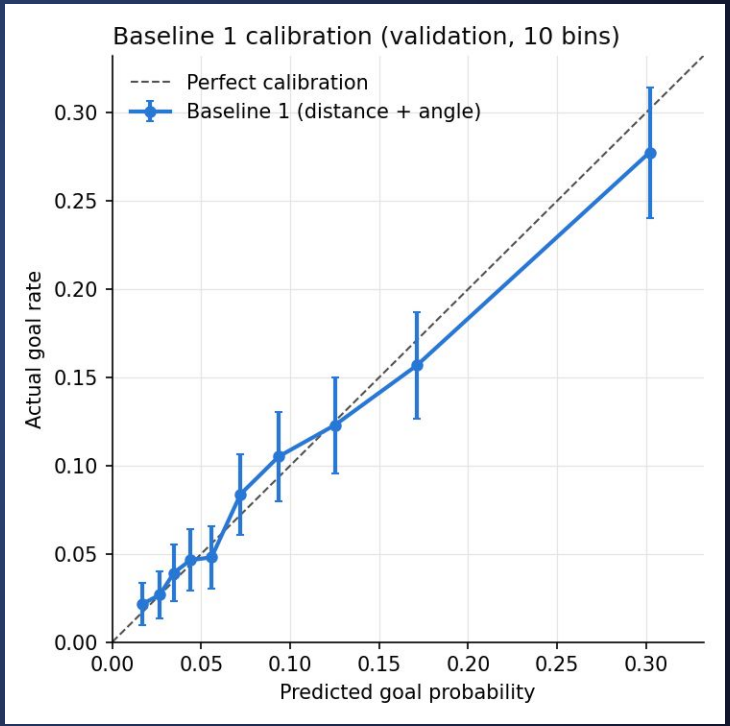
  Why M2 & M3? Baseline 1 over-predicts headers (849 predicted vs 451 actual)
- Why logistic regression: outputs a probability · standard xG baseline · coefficients → odds ratios we can explain

```
Baseline 1 coefficients:
  intercept   −1.527
  distance    −0.081
  angle       +1.146
```

Further → less likely;
wider angle → more likely.

Plan: Training & Evaluation

- Split by **match**: Train 70% · Validation 15% · Test 15%
- **Log loss:** compare models on validation
- **Calibration curve:** when the model says 20%, do ~20% score?



Next Steps

1. Fit Models 2–3 and compare on validation
2. Join football-data odds → check calibration for favourites vs underdogs
3. Final: run the chosen model on the test set once



THANKS!



Questions?

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